

DevOps Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
 - **Objective:** Share your official internship acceptance on your professional network.
 - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#devops` `#docker` `#cicd` `#progree`
- **TASK 2: Application Containerization & Asset Optimization**
 - **Objective:** Build a modular, reliable application container environment package.
 - **Requirements:** Author multi-stage configuration Dockerfiles for a multi-dependency web app runtime stack. Minimize final image layer footprints, map container environment variable configurations securely, and define functional container port routing maps.
- **TASK 3: Multi-Stage Automated CI/CD Deployment Pipeline**
 - **Objective:** Script an automated testing and integration workflow running on remote pushes.
 - **Requirements:** Configure robust automation runner scripts (GitHub Actions, GitLab CI, or Jenkins). Define pipeline blocks that automatically pull code, execute static application linter steps, run unit-testing suites, and log deployment output status metrics on execution boards.
- **TASK 4: Mini Project - Automated Infrastructure as Code & Orchestration Architecture**
 - **Objective:** Provision and orchestrate an elastic, highly available microservices cluster environment.
 - **Requirements:** Map cloud network resources programmatically using declarative Infrastructure as Code tools (Terraform or CloudFormation) to spin up isolated networks. Deploy your containerized application stack across a Kubernetes (EKS/GKE/Minikube) cluster setup, configuring persistent volume claims, Ingress controller endpoints, and Horizontal Pod Autoscalers (HPA).